AP PRECALCULUS

Unit 1 • Worksheet 1.3A • Answer key

Worksheet 1.3A • Rates of Change in Linear and Quadratic Functions

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Each list gives outputs at $x = 0, 1, 2, 3$. Which list could come from a linear function?

(A) $3, 5, 9, 15$

(B) $4, 7, 10, 13$

(C) $1, 2, 4, 8$

(D) $2, 3, 5, 8$

Solution

Answer: (B). Constant differences of 3.

Question 2

NO CALCULATOR

Each list gives outputs at $x = 0, 1, 2, 3, 4$. Which list could come from a quadratic function?

(A) $1, 3, 9, 27, 81$

(B) $5, 8, 11, 14, 17$

(C) $0, 2, 8, 18, 32$

(D) $1, 2, 4, 7, 13$

Solution

Answer: (C). Differences $2, 6, 10, 14$; second differences $4, 4, 4$ ($y = 2x^2$).

Question 3

NO CALCULATOR

For $f(x) = ax^2 + bx + c$, the second differences of $f(0), f(1), f(2), \ldots$ are all -4. What is a?

(A) -4

(B) 4

(C) -2

(D) 2

Solution

Answer: (C). Second difference $= 2a$ for step 1.

Question 4

NO CALCULATOR

The inputs in a table are $0, 2, 4, 6$ and the outputs of a quadratic function have second differences equal to 8. What is a?

(A) 1
(B) 2
(C) 4
(D) 8

Solution
Answer: (A). With step $h = 2$, second differences equal $2ah^2 = 8a$. So $8a = 8$ and $a = 1$.

Question 5 (Free response)

NO CALCULATOR

x	-1	0	1	2	3	4
y	9	4	1	0	1	4

(a) Show that the data can be modeled by a quadratic function.
(b) Find the quadratic function.
(c) On which interval is the function decreasing? How do the first differences show this?

Solution
(a) First differences $-5, -3, -1, 1, 3$; second differences all 2.
(b) $2a = 2$, $a = 1$; vertex $(2, 0)$: $y = (x - 2)^2$.
(c) Decreasing on $(-\infty, 2)$: the first differences are negative there.